Lección 4

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Plan de trabajo

Teoría de relatividad

– Cinemática Relativista

... Conservación de momento y energía

... Dispersión de partículas. Invariantes (variables) de Mandelstam

– Aplicaciones: ... efecto de Doppler relativista

► Nuestro Syllabus:

UNIDAD 1. Teoría de la Relatividad Especial
1.1. Transformaciones de Galileo
1.2. Experimento de Michelson y Morley
1.3. Postulados de Einstein
1.4. Transformaciones de Lorentz, Grupo de Lorentz
1.5. Espacio-tiempo de Minkowski y vectores 4-dimensionales
1.6. Cinemática Relativista
1.7. Dinámica Relativista
1.8. Aplicaciones: la paradoja de los gemelos, efecto Doppler relativista

Recap: Lorentz group and Einstein's velocity addition

Boosts in arbitrary direction:

$$\boldsymbol{x}' = \boldsymbol{x} + \boldsymbol{v}\left[\frac{\boldsymbol{x}\cdot\boldsymbol{v}}{v^2}(\gamma-1) - \gamma t\right], \quad t' = \gamma\left(t - \frac{\boldsymbol{x}\cdot\boldsymbol{v}}{c^2}\right)$$

Can rewrite in **compact matrix form**: $x'^{\mu} = \Lambda^{\mu}{}_{\nu}(\boldsymbol{v})\, x^{\nu}$

$$\Lambda^{\mathrm{T}}\cdot\Lambda = \Lambda\cdot\Lambda^{\mathrm{T}} = 1$$

►In general matrices Λ^{μ}_{ν} might include a superposition of Lorentz boost & rotation

Einstein's velocity addition & covariance:

$$\boldsymbol{v} = \frac{d\boldsymbol{x}}{dt} = \frac{\boldsymbol{v}' + \boldsymbol{V}\left[\frac{\boldsymbol{v}'\cdot\boldsymbol{V}}{V^2}(\gamma-1)+\gamma\right]}{\gamma\left(1+\frac{\boldsymbol{v}'\cdot\boldsymbol{V}}{c^2}\right)} \qquad (1)$$

–Guarantees that $c = \mathrm{const}$ in any frame

–Since (1) is so sophisticated/nonlinear, it is very difficult to use it to get covariant equations of motion

*Define instead **covariant** velocities

$$u^{\mu} \equiv \frac{dx^{\mu}}{d\tau} = \left(\frac{c}{\sqrt{1-v^2/c^2}}, \frac{\vec{\boldsymbol{v}}}{\sqrt{1-v^2/c^2}}\right), \quad d\tau = dt\sqrt{1-v^2/c^2}, \quad u_{\mu}u^{\mu} = c^2$$

Transformation of u^{μ} is covariant (as 4-vector $x^{\mu} = (ct, \boldsymbol{x})$): $u^{\mu} \to \Lambda^{\mu}{}_{\nu}u^{\nu}$

Proper 4-momentum and 4-force

4-momentum (definition)

$$p^\mu \equiv m\, u^\mu = \left(\frac{mc}{\sqrt{1 - v^2/c^2}}, \frac{m\vec{\boldsymbol{v}}}{\sqrt{1 - v^2/c^2}} \right),$$

- By construction the components of the 4-vector p^μ transform as coordinates x^μ and 4-velocity u^μ, viz

$$p'^\mu = \Lambda^\mu{}_\nu p^\nu$$

$$\boldsymbol{p}' = \boldsymbol{p} + \boldsymbol{v} \left[\frac{\boldsymbol{p} \cdot \boldsymbol{v}}{v^2} (\gamma - 1) - \gamma \frac{E}{c^2} \right], \quad E' = \gamma \left(E - \boldsymbol{p} \cdot \boldsymbol{v} \right)$$

4-vector of force

$$g^\mu = dp^\mu / d\tau = \left(\frac{\boldsymbol{f} \cdot \boldsymbol{v}}{c\sqrt{1 - v^2/c^2}}, \frac{\boldsymbol{f}}{\sqrt{1 - v^2/c^2}} \right), \qquad \boldsymbol{f} = d\vec{\boldsymbol{p}}/dt$$

$$\text{orthogonality}: \qquad u \cdot g \equiv u_\mu g^\mu \equiv \eta_{\mu\nu} u^\mu g^\nu = 0$$

.

Example: Motion in uniform electric or magnetic fields

As You may remember, for nonrelativistic particle:

– In homogeneous electric field, trajectory is parabola

– In homogeneous magnetic field, trajectory is a circular helix (or circumference in special llimit)

► Now we are going to consider what happens for <u>relativistic</u> particle ...

Problem 1

A relativistic particle of mass m and electric charge q moves in a constant homogeneous electric field $\boldsymbol{E}$. The initial velocity of particle $\boldsymbol{v}$ forms angle α with direction of electric field $\boldsymbol{E}$. Find the trajectory of the particle.

Problem 1

$$\frac{d\boldsymbol{p}}{dt} = q\left(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\right)$$

- Case $\boldsymbol{B} = 0$, select $\hat{\boldsymbol{x}}$ in direction of $\boldsymbol{E}$

$$\dot{p}_x = q\,E,\ \dot{p}_y = 0$$

$$p_x = q\,E\,t + p_{0x},\ \ p_y = p_{0y}$$

Can solve algebraically to fix v_x, v_y from

$$\boldsymbol{v} = \frac{d\boldsymbol{x}}{dt} = \frac{\boldsymbol{p}\,c^2}{E_{\mathrm{kin}}}$$

Problem 1

$$v_x = c^2 \frac{qEt + p_{0x}}{\sqrt{p_{0y}^2 + m^2c^2 + (qEt + p_{0x})^2}} \Rightarrow x(t) = \frac{1}{q\,E}\sqrt{\mathcal{E}_0^2 + (qEt + p_{0x})^2}$$

-hyperbolic motion in x-direction

$$v_y = c^2 \frac{p_{0y}}{\sqrt{p_{0y}^2 + m^2c^2 + (qEt + p_{0x})^2}} \Rightarrow y(t) = \frac{p_{0y}\,c}{q\,E}\,\mathrm{arcsinh}\left(\frac{qEt + p_{0x}}{\sqrt{p_{0y}^2 + m^2c^2}}\right)$$

Trajectory of particle=catenary curve

$$x(y) = \frac{\mathcal{E}_0}{q\,E}\cosh\left(\frac{qEy}{p_{0y}\,c}\right)$$

-different from parabola in classical theory

Problem 2

A relativistic particle of mass m and electric charge q moves in a constant homogeneous magnetic field $\boldsymbol{B}$. Find the trajectory of the particle.

Motion in uniform magnetic field

$$\frac{d\boldsymbol{p}}{dt} = q\left(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\right)$$
$$\boldsymbol{E} = -\boldsymbol{\nabla}\phi - \frac{\partial \boldsymbol{A}}{\partial t},$$
$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}$$

- Have $\boldsymbol{E} = 0$, select $\hat{\boldsymbol{z}}$ in direction of $\boldsymbol{B}$

$$\dot{p}_x = q\, v_y B, \quad (1)$$
$$\dot{p}_y = -q\, v_x B, \quad (2)$$
$$\dot{p}_z = 0, \quad (3)$$

NOTE: $E_{\mathrm{kin}} = \mathrm{const}$ (since $\boldsymbol{f} \cdot \boldsymbol{v} = 0$) and $\boldsymbol{p} = E_{\mathrm{kin}}\boldsymbol{v}/c^2$

$$\dot{v}_x = v_y q\, c^2 B/E_{\mathrm{kin}}, \quad (1')$$
$$\dot{v}_y = -v_x q\, c^2 B/E_{\mathrm{kin}}, \quad (2')$$
$$\dot{v}_z = 0, \quad (3) \Rightarrow v_z = v_{0z}$$

Motion in uniform magnetic field

Evaluate $(1') + i\,(2')$:

$$\dot{\mathcal{V}} = -i\,\mathcal{V}\,\Omega$$

where $\Omega \equiv q\,c^2 B/E_{\rm kin}$ and $\mathcal{V} = v_x + i\,v_y$

$$\Rightarrow \mathcal{V} = \mathcal{V}_0 e^{-i\Omega t}$$

$$\Rightarrow x + iy = \frac{i\mathcal{V}_0}{\Omega} e^{-i\Omega t} \quad (3)$$

$$v_x = \Re e\,(\mathcal{V})\,, \quad v_y = \Im m\,(\mathcal{V})$$

- Similar to nonrelativistic case, particle's trajectory is helicoidal line in general (circumference if $v_{0z} = 0$)
- Angular frequency in transverse plane

$$\Omega = q\,c^2 B/E_{\rm kin} = \frac{q\,B}{m}\sqrt{1 - v^2/c^2}$$

- Radius of circumference in transverse plain from (3):

$$R = v_\perp/\Omega = \frac{m v_\perp}{q\,B\sqrt{1 - v^2/c^2}}$$

Similarly integration is done for other cases, see e.g. sections §19-23 in Landau for more details

Caution with relativistic system

- Relativistic system is not just modification of energy/momentum

►some qualitative features are *different* from $v \ll c$ limit

- We no longer have long-distance interaction

– no $U(\boldsymbol{r}_1 - \boldsymbol{r}_2)$-type terms are allowed in the action

–For closed system should consider interaction *in the same point* x^μ, field+particle.

– MUST include terms both for partciles and fields to keep covariance

►Field (e.g. electrogamgenetic) carries away part of energy-momentum, will affect balance of energy-momentum

Can't introduce concept of center-of-mass

$$\boldsymbol{R}_{\mathrm{CM}}(t) = \frac{\sum_{i=1}^{N} m_i \, \boldsymbol{x}_i(t)}{\sum_{i=1}^{N} m_i}$$

and "separate" its motion

►Collective momentum $\boldsymbol{P} = \sum \boldsymbol{p}_i$ has nothing to do with $d\boldsymbol{R}_{\mathrm{CM}}(t)/dt$.

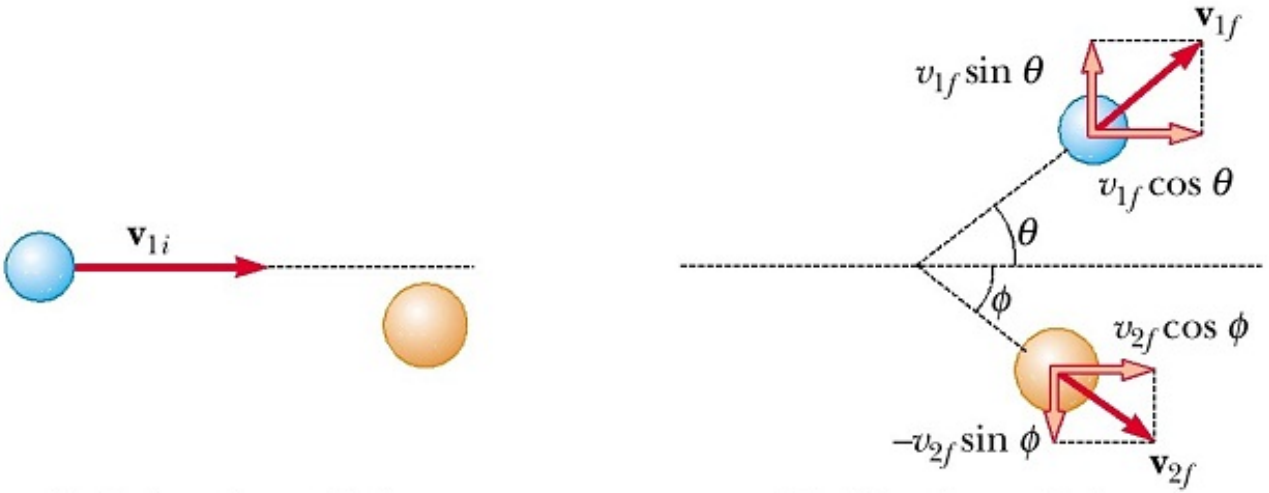
►We can choose "C-frame" in which $\boldsymbol{P} = 0$, but it is not CM frame!

Energy-momentum conservation

- *If we have a system of particles, then we have energy-momentum conservation, as expected from Noether theorem. However, we should use* <u>*relativistic*</u> *expressions for them*
- *We may write energy-momentum conservation in relativistic notations, using 4-vectors as*

$$\sum_n p^\mu_{(n)} = \text{const}$$

- *In order to illustrate how this works, we'll consider elastic collision* $2 \to 2$.



(a) Before the collision (b) After the collision

Decay of a particle

Assume that a particle of mass M decays into two fragments of masses m_1 and m_2 (obviously, $M > m_1 + m_2$).

1) Find the energies of the particles in the rest frame of the initial particle ("C-frame").
2) Assume that the initial particle moves in the laboratory with velocity V, and after the decay, the fragments follow straight lines which form angles θ_1, θ_2 with trajectory of the initial particle. Evaluate the relation between angles θ_1, θ_2 and the energies of the particles

Decay of a particle (solution)

Energy:momentum conservation:

$$M\,c^2 = E_1 + E_2, \ (1) \qquad 0 = \boldsymbol{p}_1 + \boldsymbol{p}_2 \ (2) \qquad \boldsymbol{p}_2 = -\boldsymbol{p}_2 \ (2')$$

$$E_{1,2} = c\,\sqrt{p_{1,2}^2 + m_{1,2}^2 c^2}$$

$$\Rightarrow \frac{E_1^2}{c^2} - m_1^2 c^2 = \frac{E_2^2}{c^2} - m_2^2 c^2 \quad (3)$$

$$\Rightarrow \frac{E_1^2 - E_2^2}{c^2} = \frac{(E_1 - E_2)\,(E_1 + E_2)}{c^2} = \left(m_1^2 - m_2^2\right) c^2 \quad (3')$$

From (1) and (3')

$$\Rightarrow E_1 = \frac{M^2 + m_1^2 - m_2^2}{2\,M}, \qquad E_2 = \frac{M^2 - m_1^2 + m_2^2}{2\,M}$$

Decay of a particle (solution)

We know that in C-frame and lab-frame the energies are bound by

$$E_1^{(\mathrm{C})} = \frac{E_1^{(\mathrm{lab})} - V\, p^{(\mathrm{lab})} \cos\theta}{\sqrt{1 - V^2/c^2}}$$

where $E_1^{(\mathrm{C})}$ was found earlier, depends on $m_{1,2}$ only.

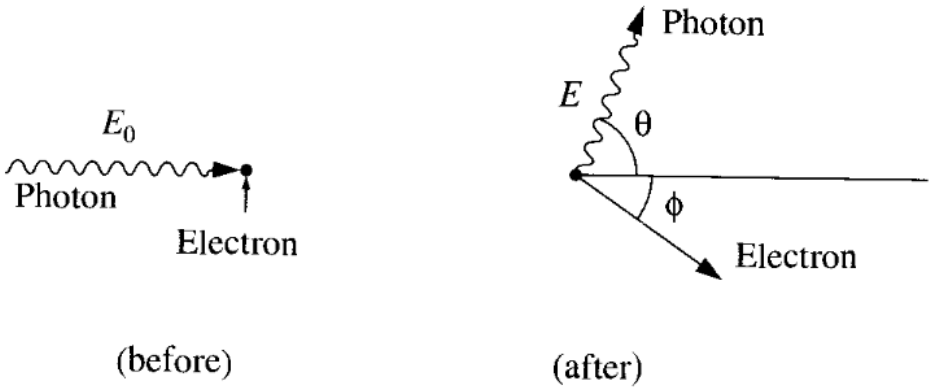
$$E_1^{(\mathrm{lab})} = c\sqrt{\left(p^{(\mathrm{lab})}\right)^2 + m_1^2 c^2},$$

$$\Rightarrow \cos\theta = \frac{E_1^{(\mathrm{lab})} - E_1^{(\mathrm{C})}\sqrt{1 - V^2/c^2}}{V\, p^{(\mathrm{lab})}} = \frac{E_1^{(\mathrm{lab})} - E_1^{(\mathrm{C})}\sqrt{1 - V^2/c^2}}{V\sqrt{\left(E_1^{(\mathrm{lab})}/c\right)^2 - m_1^2 c^2}}$$

Can solve algebraically for $E_1^{(\mathrm{lab})}$ for a given θ_1. Similarly for the second particle

Compton scattering

Photon with energy E_0 scatters on electron, initially at rest. Find the relation between the energy of the scattered photon and the scattering angle θ



Caution:

- In *classical* electrodynamics we work with EM fields $\boldsymbol{E}$, $\boldsymbol{B}$
- Concept of photon appears in *quantum* electrodynamics. Now consider it simply as *ultrarelativistic particle* with energy-mass relation $E = |\boldsymbol{p}|\, c$ (moving with speed of light c).
- Relation to the wavelength of EM wave:

$$\lambda := \frac{h\, c}{E}$$

Compton scattering

►We'll work in lab-frame, use energy-momentum conservation

–Use notations p_e, p_p for momenta of scattered electron and photon

After scattering:

$$p_0 = p_e \cos\phi + p_p \cos\theta \quad (1)$$

$$p_e \sin\phi = p_p \sin\theta \quad (2)$$

$$p_0 c + mc^2 = p_p c + c\sqrt{p_e^2 + m^2c^2} \quad (3)$$

-system of 3 equations with unknown $p_p = |E_e|/c$, ϕ, p_e, use algebraic methods to solve:

$$p_0 - p_p \cos\theta = p_e \cos\phi \quad (1')$$

$$\Rightarrow (p_0 - p_p - mc)^2 = p_e^2 + m^2c^2 \quad (3')$$

$(1')^2 + (2)^2$:

$$(p_0 - p_p \cos\theta)^2 + p_p^2 \sin^2\theta = p_e^2$$

$$p_0^2 + p_p^2 - 2p_0 p_p \cos\theta = p_e^2 \quad (4)$$

$(4) - (3')$:

$$2p_0 p_p (1 - \cos\theta) + 2mc(p_0 - p_p) = 2m^2c^2$$

Take into account $E_p = p_p c$, $E_0 = p_0 c$ and solve equation:

$$E_p = p_p c = \frac{1}{(1 - \cos\theta)/mc^2 + 1/E_0}$$

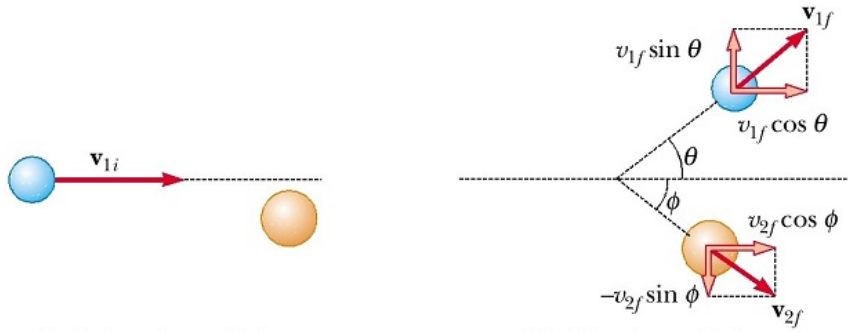
$$\lambda = \lambda_0 + \underbrace{\frac{h}{mc}}_{\lambda_e}(1 - \cos\theta)$$

λ_e-wavelength of electron

Nonrelativistic elastic scattering

(***Classical mechanics***)

►Consider $2 \to 2$ elastic scattering



(a) Before the collision (b) After the collision

- Energy-momentum conservation: 4 independent constraints
  $\Rightarrow$if we know ($\boldsymbol{p}_{1\,I}$, $\boldsymbol{p}_{2\,I}$), and only one of the angles (θ_1, θ_2) or energies *after* scattering (E_1, E_2), can reconstruct the remaining variables

Center of mass frame

1) The scattering always happens in a plane
2) The absolute values of momenta are conserved, $|\boldsymbol{p}_{1\,I}| = |\boldsymbol{p}_{1\,F}| = |\boldsymbol{p}_{2\,I}| = |\boldsymbol{p}_{2\,F}|$
3) The scattering angles are equal, $\theta_1 = \theta_2$
4) The relative motion:

$$\boldsymbol{r}_{12} = \boldsymbol{r}_1 - \boldsymbol{r}_2$$

with reduced mass
$\mu = m_1 m_2/(m_1 + m_2)$

Relativistic elastic scattering

In **relativistic case** there are also relations dictated by energy-momentum conservation. We'll derive them, and will use only covariant notations (in order to get covariant results):

In relativistic case angles, energies transform using expression which are significantly more complicated than in non-relativistic case

It might be easier/faster to formulate possible kinematic constraints using invariant <u>*scalar products of 4-vectors*</u> $p_a \cdot p_b \equiv \eta_{\mu\nu} p_a^\mu p_b^\nu$, *and later just write out these 4-momenta in terms of the angles/energies in the frame which is of interest for us.*

Relativistic elastic scattering

►4-momentum conservation:

$$p^{\mu}_{1i} + p^{\mu}_{2i} = p^{\mu}_{1f} + p^{\mu}_{2f} \quad (1)$$

-4 equations

►"Onshell conditions" $p^2_{1,2} = m^2_{1,2}c^2$ $(p^2 \equiv E^2/c^2 - p^2)$

– ... should write for all 4 momenta $(p_{1i,\,2i,\,1f,\,2f})$

►Numerous relations:

$$(p_{1i} + p_{2i})^2 = (p_{1f} + p_{2f})^2$$

$$\Rightarrow \underbrace{p^2_{1i}}_{m^2_1} + \underbrace{p^2_{2i}}_{m^2_2} + 2p_{1i} \cdot p_{2i} = \underbrace{p^2_{1f}}_{m^2_1} + \underbrace{p^2_{2f}}_{m^2_2} + 2p_{1f} \cdot p_{2f}$$

$$\Rightarrow p_{1i} \cdot p_{2i} = p_{1f} \cdot p_{2f}$$

$$(p_{1f} - p_{1i})^2 = (p_{2f} - p_{2i})^2$$

$$\Rightarrow \underbrace{p^2_{1f}}_{m^2_1} + \underbrace{p^2_{1i}}_{m^2_1} - 2p_{1f} \cdot p_{1i} = \underbrace{p^2_{2f}}_{m^2_2} + \underbrace{p^2_{2i}}_{m^2_2} - 2p_{2f} \cdot p_{2i}$$

$$\Rightarrow m^2_1 - p_{1f} \cdot p_{1i} = m^2_2 - p_{2f} \cdot p_{2i}$$

Relativistic elastic scattering

- 4-momentum conservation:

$$p_{1i}^{\mu} + p_{2i}^{\mu} = p_{1f}^{\mu} + p_{2f}^{\mu} \quad (1)$$

- Consequences:

$$(p_{1f} - p_{2i})^2 = (p_{2f} - p_{1i})^2$$
$$\Rightarrow p_{1f} \cdot p_{2i} = p_{2f} \cdot p_{1i}$$

$$p_{2f}^2 = (p_{1i} + p_{2i} - p_{1f})^2$$
$$\Rightarrow \cancel{m_2^2} = 2\, m_1^2 + \cancel{m_2^2} + 2p_{1i} \cdot p_{2i}$$
$$- 2p_{1i} \cdot p_{1f} - 2p_{2i} \cdot p_{1f}$$

$$\Rightarrow p_{1i} \cdot p_{2i} + m_1^2 = p_{1i} \cdot p_{1f} + p_{2i} \cdot p_{1f}$$

substitution ($i \leftrightarrow f$):

$$\Rightarrow p_{1f} \cdot p_{2f} + m_1^2 = p_{1i} \cdot p_{1f} + p_{1i} \cdot p_{2f}$$

substitution ($1 \leftrightarrow 2$):

$$\Rightarrow p_{1i} \cdot p_{2i} + m_2^2 = p_{2i} \cdot p_{2f} + p_{1i} \cdot p_{2f}$$

substitution ($1 \leftrightarrow 2$, $i \leftrightarrow f$):

$$\Rightarrow p_{1f} \cdot p_{2f} + m_2^2 = p_{2i} \cdot p_{2f} + p_{1i} \cdot p_{2f}$$

Relativistic elastic scattering

$$p_{1i} \cdot p_{2i} = p_{1f} \cdot p_{2f} \quad (2)$$

$$m_1^2 - p_{1f} \cdot p_{1i} = m_2^2 - p_{2f} \cdot p_{2i} \quad (3)$$

$$p_{1f} \cdot p_{2i} = p_{2f} \cdot p_{1i} \quad (4)$$

$$p_{1i} \cdot p_{2i} + m_1^2 = p_{1i} \cdot p_{1f} + p_{2i} \cdot p_{1f} \quad (5)$$

$$p_{1f} \cdot p_{2f} + m_1^2 = p_{1i} \cdot p_{1f} + p_{1i} \cdot p_{2f} \quad (6)$$

$$p_{1i} \cdot p_{2i} + m_2^2 = p_{2i} \cdot p_{2f} + p_{1i} \cdot p_{2f} \quad (7)$$

- Let's assume one of the particles is at rest, $p_{2i} = (m_2 c, 0)$:

$$p_{1i} = \left(\sqrt{p_1^2 + m^2 c^2}, \, p_1 \right), \quad |\boldsymbol{p}_1| = \sqrt{E_1^2/c^2 - m_1^2 c^2}$$

$$p_{1i} \cdot p_{2i} = E_{1i} \, m_2 c, \quad p_{2i} \cdot p_{1f} = m_2 c \, E_{1f}, \quad p_{1i} \cdot p_{1f} = E_{1i} E_{1f} - |\boldsymbol{p}_{1i}| \, |\boldsymbol{p}_{1f}| \cos\theta_1$$

From (5)

$$\cos\theta_1 = \frac{E_{1f}/c \, (E_{1i}/c + m_2 c)}{|\boldsymbol{p}_{1i}| \, |\boldsymbol{p}_{1f}|} - \frac{E_{1i} m_2 c + m_1^2 c^2}{|\boldsymbol{p}_{1i}| \, |\boldsymbol{p}_{1f}|}$$

Mandelstam invariants - Elastic scattering

Previous results:

- Energy of initial & scattered particles fully fix kinematics (angles)
- Note that energies/angles depend essentialy on reference frames, not very convenient
- Now we'll show that Mandelstam invariants (variables)

$$s := (p_{1i} + p_{2i})^2 = (p_{1f} + p_{2f})^2$$
$$t := (p_{1f} - p_{1i})^2 = (p_{2f} - p_{2i})^2$$
$$u := (p_{1f} - p_{2i})^2 = (p_{2f} - p_{1i})^2$$

are sufficient to fix the kinematics of the process $2 \to 2$.

Lemma

Assume that we have some **inelastic** process

$$1 + 2 \to 3 + 4.$$

Masses of all particles 1,2,3,4 are different. Then Mandelstam variables s, t, u are bound by

$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^3 \quad (A)$$

Proof:
Momentum conervation:

$$p_1 + p_2 - p_3 - p_4 = 0 \quad (1)$$

$$s + t + u = (p_1 + p_2)^2 + (p_1 - p_3)^2 + (p_1 - p_4)^2 \quad (2)$$
$$= (p_3 + p_4)^2 + (p_2 - p_4)^2 + (p_2 - p_3)^2 \quad (3)$$

– Let's sum (2+3) and take into account that $p_i^2 = m_i^2$, as well as identities

$$3p_i^2 = 2m_i^2 + p_i \cdot p_i$$
$$2p_i \cdot p_j = p_i \cdot p_j + p_j \cdot p_i$$

Lemma

$$
\begin{aligned}
&2\left(s+t+u\right)= \\
&=2\left(m_1^2+m_2^2+m_3^2+m_4^2\right) \\
&+p_1\cdot\left(p_1+p_2-p_3-p_4\right)+ \\
&+p_2\cdot\left(p_1+p_2-p_3-p_4\right)+ \\
&-p_3\cdot\left(p_1+p_2-p_3-p_4\right)+ \\
&-p_4\cdot\left(p_1+p_2-p_3-p_4\right)= \\
&2\left(m_1^2+m_2^2+m_3^2+m_4^2\right)
\end{aligned}
$$

$\Rightarrow$Conclude that (A) is correct, Q.E.D.

Mandelstam variables - Elastic scattering

$$p_{1i} \cdot p_{2i} + m_1^2 = p_{1i} \cdot p_{1f} + p_{2i} \cdot p_{1f} \quad (5)$$

- (invariant) Mandelstam variables:

$$s := (p_{1i} + p_{2i})^2 = (p_{1f} + p_{2f})^2$$
$$t := (p_{1f} - p_{1i})^2 = (p_{2f} - p_{2i})^2$$
$$u := (p_{1f} - p_{2i})^2 = (p_{2f} - p_{1i})^2$$

Common practice: express all physical observables/cross-sections in terms of s, t, u instead of jumping from one frame to another

Elastic scattering:

$$s + t + u = 2\left(m_1^2 + m_2^2\right)$$

$\Rightarrow$-only 2 of them are independent

Rest frame of one of the particles:

$$s = (E_{1i} + m_2)^2 - p_1^2 = m_1^2 + m_2^2 + 2\,E_{1i} m_2$$

$\Rightarrow$ s is related to energy of collision

Mandelstam variables - Elastic scattering

$$p_{1i} \cdot p_{2i} + m_1^2 = p_{1i} \cdot p_{1f} + p_{2i} \cdot p_{1f} \quad (5)$$

- (invariant) Mandelstam variables:

$$t := (p_{1f} - p_{1i})^2 = 2\left(m_1^2 - p_{1i} \cdot p_{1f}\right)$$

-"absolute value" of momentum transfer from one particle to another; from (5)

$$p_{1i} \cdot p_{1f} = E_{1i} m_2 + m_1^2 - m_2 E_{1f} = m_1^2 + m_2 \left(E_{1i} - E_{1f}\right)$$

so

$$t = -2m_2 \left(E_{1i} - E_{1f}\right) \leq 0$$

is related to energy loss of one particle. From

$$\left(m_1^2 - \frac{t}{2}\right) = p_{1i} \cdot p_{1f} = E_{1i} E_{1f} - |\boldsymbol{p}_{1i}| \, |\boldsymbol{p}_{1f}| \cos\theta_1$$

$$|\boldsymbol{p}_1| = \sqrt{E_1^2/c^2 - m_1^2 c^2}$$

can relate t to the scattering angle θ_1 of elastic scattering

Mandelstam variables-Graphic illustration

Assume that we have some inelastic process

$$1 + 2 \to 3 + 4.$$

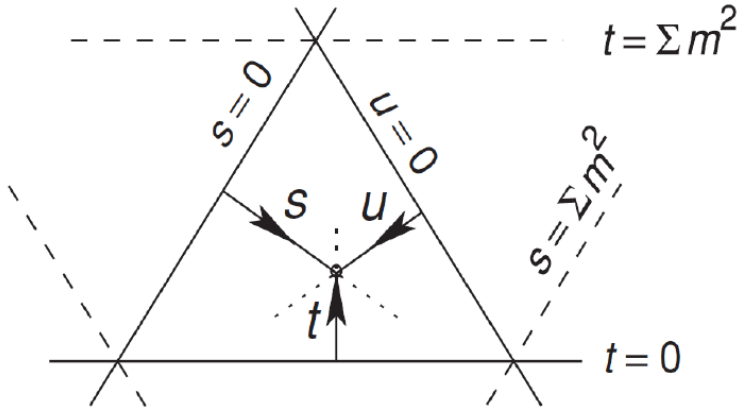
Masses of all particles 1,2,3,4 are different. Then Mandelstam variables s, t, u are bound by

$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^3 \quad (A)$$

▷Relation (A) implies that only 2 variables are independent

▷ Viviani's theorem: for any point inside equilateral triangle sum of distances equals height of triangle

⇒Graphical illustration of relation (A):



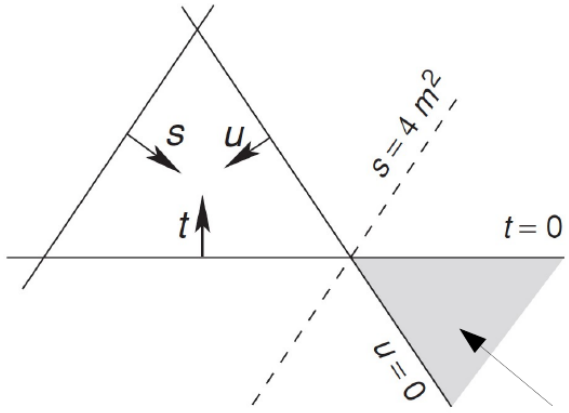
Assume for simplicity that:

▷ all masses are equal, $m_i \equiv m$

▷ work in CM frame of particles $1, 2$, so

$$p_1 = (E, \boldsymbol{p}),$$
$$p_2 = (E, -\boldsymbol{p}),$$
$$E = \sqrt{m^2 + \boldsymbol{p}^2}$$

$$s = (p_1 + p_2)^2 =$$
$$= 2m^2 + 2\left(E^2 + \boldsymbol{p}^2\right) =$$
$$= 4m^2 + 4\boldsymbol{p}^2 \geq 4m^2$$



physical region of scattering

Mandelstam variables-Exercises

Assume that we study process

$$1 + 2 \to 3 + 4.$$

Masses of all particles 1,2,3,4 are different. Prove that the Mandelstam variables s, t, u are bound by inequalities

$$s \geq (m_1 + m_2)^2 , \quad (A)$$
$$s \geq (m_3 + m_4)^2 , \quad (B)$$
$$t \leq (m_2 - m_4)^2 , \quad (C)$$
$$t \leq (m_1 - m_3)^2 , \quad (D)$$
$$u \leq (m_1 - m_4)^2 , \quad (E)$$
$$u \leq (m_2 - m_3)^2 . \quad (F)$$

Mandelstam variables-Exercises

Assume we study process

$$1 + 2 \to 3 + 4.$$

and we've chosen 3 independent momenta q_1, q_2, q_3 (the 4th might be found from energy-momentum conservation) and construct vector

$$L_\alpha = \varepsilon_{\alpha\beta\mu\nu} q_1^\beta q_2^\mu q_3^\nu .$$

Show that in the rest frame of particle $1, 2$ or 3 the vector L_α cannot have 0-component, and for this reason

$$L_\alpha L^\alpha \equiv -\boldsymbol{L}^2 \leq 0 \quad (A)$$

Show that in arbitary frame

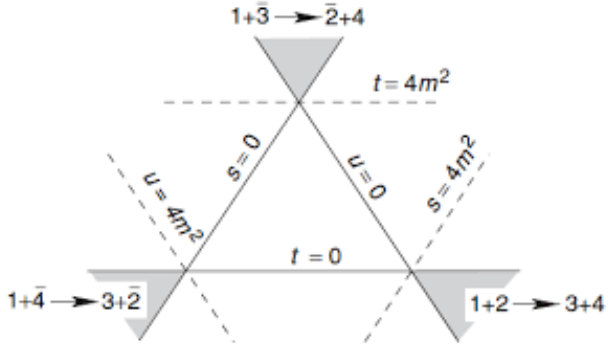
$$-L_\alpha L^\alpha = \det \begin{vmatrix} q_1^2 & q_1 \cdot q_2 & q_1 \cdot q_3 \\ q_1 \cdot q_2 & q_2^2 & q_2 \cdot q_3 \\ q_1 \cdot q_3 & q_2 \cdot q_3 & q_3^2 \end{vmatrix} = \det \begin{vmatrix} m_1^2 & \frac{s - m_1^2 - m_2^2}{2} & \frac{m_1^2 + m_3^2 - t}{2} \\ \frac{s - m_1^2 - m_2^2}{2} & m_2^2 & \frac{m_2^2 + m_3^2 - u}{2} \\ \frac{m_1^2 + m_3^2 - t}{2} & \frac{m_2^2 + m_3^2 - u}{2} & m_3^2 \end{vmatrix} \geq 0$$

-this gives nontrivial conditions for possible values of s, t, u (*Kibble*)

Mandelstam variables

Further reading:

- *Problems* in Landau, Vol. 2, Section 13
- Vol. 4 of Berestetskii, V.B., Lifshitz, E.M. and Pitaevskii, L.P. (1974) Relativistic Quantum Theory. Pergamon Press, Section 66-67
- If You will try to seek Mandelstam variables on your own, You may find something like:



– in Quantum Field Theory it is shown that the same cross-section (function) describes the process $1 + 2 \to 3 + 4$, as well as processes which might be obtained replacing particle with momentum p in initial state with its antiparticle with momentum $-p$ in final state (and vice versa). This leads to permutation of variables s, t, u and explains why we get into "unphysical" region (In the above-given plot $\bar{a}$ implies antiparticle of particle a).

Relativistic Doppler effect

We've discussed several examples how the relativistic effects manifest themselves, and in your homeworks ("Tareas") you will analyze more complicated systems

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There is just one more topic - "Doppler effect" which we need to discuss and which is mentioned in our syllabus, so need to discuss it

– Please try to recall from undergraduate courses what is Doppler effect ?
  * Should we move with relativistic velocities to see it ?

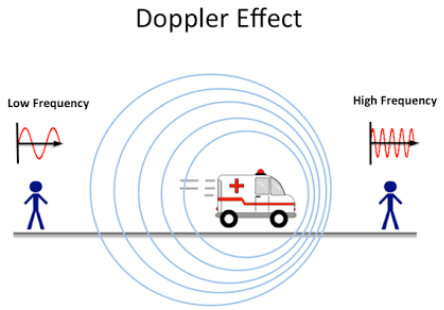
Relativistic Doppler effect

We've discussed several examples how the relativistic effects manifest themselves, and in your homeworks ("Tareas") you will analyze more complicated systems

There is just one more effect which we need to discuss and which is mentioned in our syllabus, so need to discuss it

- The Doppler effect is a change of observed frequency when the observer and the source are moving inside the medium

►Exists in classical mechanics (e.g. acoustics):



$$\omega_{\text{observer}} = \omega_{\text{source}} \left(\frac{1 - \frac{\boldsymbol{v}_o \cdot \boldsymbol{n}}{u}}{1 - \frac{\boldsymbol{v}_s \cdot \boldsymbol{n}}{u}} \right)$$

▷$\vec{\boldsymbol{v}}_s$ and $\vec{\boldsymbol{v}}_o$ are velocities of the source and the observer with respect to the medium

▷ $\boldsymbol{n}$ is the direction from source to observer, $n^2 = 1$

▷ u is the speed of wave in the medium (e.g. acoustic wave)

►This effect is NOT related to Lorentz time dilation!

Relativistic Doppler effect

- In electromagnetism we do not have "medium" (ether), all reference frames are equivalent, so only the relative motion matters.

►Doppler effect: change of frequency (wavelength) due to relative motion of the source/detector.

–Similar to nonrelativistic case, have linear term

$$\left(\frac{\Delta\omega}{\omega}\right)_{\text{nonrel}} \sim \frac{|v_i|}{c}$$

(due to motion, change time between arrival of the surfaces of constant phase)

–Additional contribution: Lorentz time dilation effects,

$$\Delta t_{\text{lab}} = \frac{\Delta t_{\text{r.f.}}}{\sqrt{1 - v^2/c^2}} \approx \Delta t_{\text{r.f.}} \left(1 + \frac{v^2}{2c^2}\right) \Rightarrow \left(\frac{\Delta\omega}{\omega}\right)_{\text{rel}} \sim \frac{v^2}{2c^2} \ll \left(\frac{\Delta\omega}{\omega}\right)_{\text{nonrel}}$$

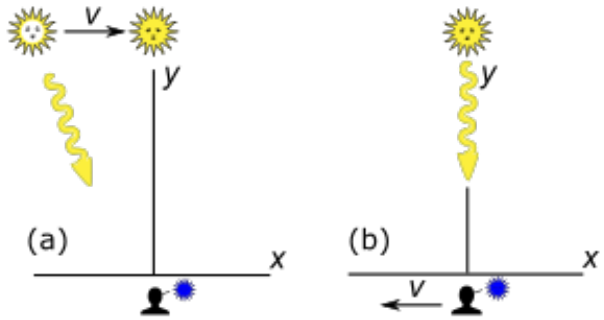
become significant when velocities $|v_i| \sim c$

►**Question:**

If relativistic Doppler is so tiny, who can we observe it/distinguish from usual (nonrelativistic)?

How to measure tiny relativistic Doppler?

- Can observe Doppler effect analyzing the shift of the spectral lines (measurable with exceptionally good precision, sensitive test of relativistic effects)
  - Nonrelativistic Doppler depends on orientation of velocity $\boldsymbol{v}$: depends on projection of $\boldsymbol{v}$ onto direction of observation
  - Relativistic Doppler does not depend on orientation of velocity $\boldsymbol{v}$



$\Rightarrow$In (a) dominates linear nonrelativistic Doppler

$\Rightarrow$In (b) linear nonrelativistic Doppler vanishes, can see relativistic Doppler

Relativistic Doppler effect (mathematics)

Let's assume that a source emits monochromatic plane wave in its rest frame:

$$\begin{pmatrix} \boldsymbol{E} \\ \boldsymbol{B} \end{pmatrix} = \begin{pmatrix} \boldsymbol{E}_0 \\ \boldsymbol{B}_0 \end{pmatrix} e^{i(\omega t - \boldsymbol{k}\cdot\boldsymbol{x})}, \quad \omega = |\boldsymbol{k}|\ c,$$

$$(\omega t - \boldsymbol{k}\cdot\boldsymbol{x}) = k^\mu x_\mu, \quad k^\mu = \left(\frac{\omega}{c},\ \boldsymbol{k}\right).$$

– For a moment ignore $\boldsymbol{E}_0, \boldsymbol{B}_0$ which determine polarization
– Plane wave maintains its form in any reference frame, so k^μ MUST transform as components of 4-vector

$$\omega' = \frac{\omega - V\,k_x}{\sqrt{1 - V^2/c^2}}, \quad k'_x = \frac{k_x - V\,\omega/c^2}{\sqrt{1 - V^2/c^2}}, \quad k'_y = k_y, \quad k'_z = k_z,$$

$$\omega = \frac{\omega' + V\,k'_x}{\sqrt{1 - V^2/c^2}}, \quad k_x = \frac{k_x + V\,\omega/c^2}{\sqrt{1 - V^2/c^2}}, \quad k'_y = k_y, \quad k'_z = k_z.$$

* Can see that even for $k'_x = 0$ (emission in transverse direction) the wave changes its frequency:

$$\omega = \frac{\omega'}{\sqrt{1 - V^2/c^2}} \approx \omega' \left(1 + \frac{V^2}{c^2}\right) \geq \omega'$$

This is manifestation of 2nd order Doppler effect due to relativistic correction

Relativistic Doppler effect (mathematics)

Generalization of our result for arbitrary direction of $\boldsymbol{v}$

$$k_0' = \gamma\left(k_0 - \frac{\boldsymbol{v}\cdot\boldsymbol{k}}{c}\right),$$
$$k_{||}' = \gamma\left(k_{||} - \frac{v}{c}k_0\right),\ \boldsymbol{k}_\perp' = \boldsymbol{k}_\perp$$

Recall that $\omega = c|\boldsymbol{k}|$, so can get for transformation of frequencies (Doppler effect) and directions of light rays

$$\Rightarrow \omega' = \gamma\omega\left(1 - \frac{v}{c}\cos\theta\right),$$
$$\Rightarrow \tan\theta' = \frac{\sin\theta}{\gamma(\cos\theta - v/c)},$$

(θ, θ' are measured w.r.t. direction of $\boldsymbol{v}$)

– If monochromatic source with frequency ω emits isotropically in its own reference frame, in any other frame observed frequency will depend on direction (angle θ)

– Note that for $\theta = \pi/2$ get $\omega_\perp = \omega\gamma > \omega$, transverse Doppler shift (merely relativistic effect, $\sim v^2/c^2$).

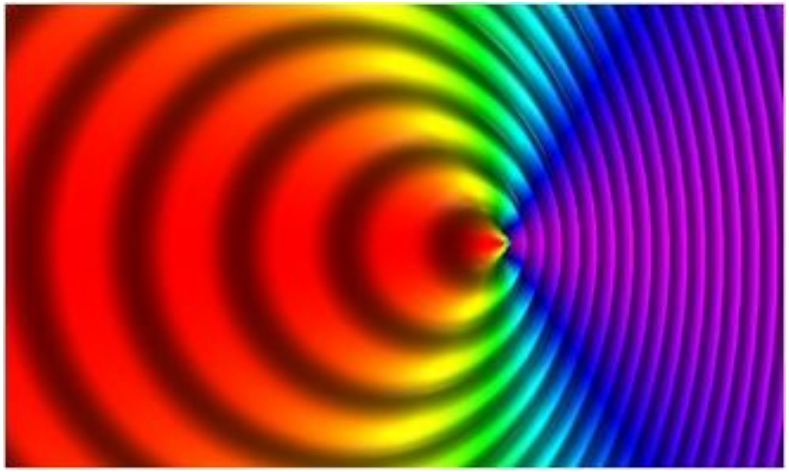
Relativistic Doppler effect (mathematics)

– If monochromatic source with frequency ω' emits isotropically in its own reference frame, in any other frame observed frequency will depend on direction (angle θ)

$$\Rightarrow \omega = \frac{\omega' \sqrt{1 - v^2/c^2}}{\left(1 - \frac{v}{c}\cos\theta\right)}$$

As the source passes by, the angle θ and frequency ω change constantly from $\omega > \omega'$ at $\theta \ll \pi/2$ (blushift) up to $\omega < \omega'$ for $\theta \gg \pi/2$ (redshift)

Nice visualization from Wikipedia:



► (Green) source moves to the right

– Observer sees the spectrum blue-shifted, if the source approaches to him

– Observer sees the spectrum red-shifted, if the source moves away from him.

Block 1 - Closing remark

We've covered all the topics of block 1.

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Please let me know if You have any questions?